

Set 1: Monthly Sales Data

Dataset: Monthly Sales

- 1. Create a **line chart** to visualize the monthly sales. Label the axes and title the chart appropriately.
- 2. Generate a **bar chart** to display the top-selling products for the year. Label the chart elements.
- 3. Develop a **scatter plot** to explore the relationship between advertising budget and monthly sales. Explain the insights drawn from the scatter plot.
- 4. Build an **interactive dashboard** combining the line chart and bar chart to allow users to explore sales data interactively.

Month	Sales (in \$)
January	15,000
February	18,000
March	22,000
April	20,000
May	23,000

SET 2.Customer Feedback Analysis

Dataset: Customer Satisfaction

Customer ID	Age	Satisfaction Score
1	25	4
2	30	5
3	35	3
4	28	4
5	40	5

Tasks

- 1. Create a **histogram** to represent the distribution of customer ages. Label the axes and title the chart.
- 2. Generate a **pie chart** to display the overall distribution of customer satisfaction scores. Include appropriate labels.
- 3. Build a **stacked bar chart** to visualize the distribution of customer satisfaction scores by age group.
- 4. Develop a **word cloud** from open-ended customer feedback to identify prevalent customer sentiments.

Set 3: Employee Performance Evaluation

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Tasks

1. Create a **line chart** to visualize the performance trend of employees over time. Include a legend and labels.
2. Generate a **bar chart** showing the distribution of employees across different departments. Label the chart elements.
3. Build a **scatter plot** to analyze the correlation between years of service and performance scores. Explain any insights.
4. Develop a **Tableau dashboard** with the line chart and bar chart for interactive exploration.

Set 4: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

1. **Create a bar chart** to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. **Generate a stacked bar chart** to show the quantity of each product within different product categories.
3. **Build a scatter plot** to explore the relationship between product price and quantity available. Explain the findings.
4. **Develop a Tableau dashboard** with the bar chart and stacked bar chart to allow users to interact with the data.

Set 5: Website Analytics

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

1. **Create a line chart** to visualize the trend in daily page views over time. Label the axes and title the chart.
2. **Generate a bar chart** showing the top **N** days with the highest click-through rates. Label the chart elements.
3. **Develop a stacked area chart** to display the distribution of user interactions (**likes, shares, comments**) on a website.
4. **Build a Tableau dashboard** with the line chart and bar chart for interactive exploration of website traffic data.

Set 6: Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

1. **Create a grouped bar chart** to visualize the sales of each product for the first quarter. Label the chart elements.
2. **Generate a stacked area chart** to represent the overall sales trend for all products over the three months.
3. **Build a table** to show the monthly sales data for each product. Label the table elements.
4. **Develop a Tableau dashboard** combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

Set 7: Customer Demographics Analysis

Dataset: Customer Demographics

Customer ID	Age	Gender	Income (in \$)
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

1. **Create a bar chart** to visualize the distribution of customer ages. Label the axes and title the chart.
2. **Generate a pie chart** to represent the distribution of customers by gender.
3. **Build a table** to show the demographic information for each customer. Label the table elements.
4. **Develop a Tableau dashboard** combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

Set 8: Online Learning Activity Analysis

Dataset: Online_Learning_Activity.csv

Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched	Quiz_Score	Login_Date
L01	Male	20	R	3.5	12	78	2025-01-05
L02	Female	22	R	4.2	15	85	2025-01-05
L03	Male	19	SQL	2.0	8	65	2025-02-08
L04	Female	21	R	5.0	18	92	2025-02-08
L05	Male	23	R	2.5	9	70	2025-03-12
L06	Female	20	SQL	4.0	14	88	2025-03-12

1. **Import the dataset in R.** Plot a histogram for **Quiz_Score**. Draw a boxplot of **Quiz_Score** by **Course**. Add title and labels. Interpret student performance.
2. **Create a scatter plot using R programming** between **Study_Time** and **Quiz_Score**. Represent **Videos_Watched** using bubble size. Apply transparency. Explain the learning pattern.

3. **Using R programming**, convert **Login_Date** into **Date** format. Calculate the average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.
4. **Import Online_Learning_Activity.csv into Tableau**. Create a bar chart showing average quiz score by **Course**. Create a scatter plot (**Study_Time vs Quiz_Score**). Add filters for **Gender** and **Course**. Design a mini interactive dashboard.

Set 9: Survey Responses Analysis

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

1. **Create a grouped bar chart** to visualize the distribution of answers for **Question 1**. Label the chart elements.
2. **Generate a stacked bar chart** to represent the overall distribution of responses for all three questions.
3. **Build a table** to show the survey response data for each survey. Label the table elements.
4. **Develop a Tableau dashboard** combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Set 10

Product Category Analysis

Dataset: Product Categories

Category	Sales (in \$)
Electronics	50000
Clothing	35000
Appliances	40000

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

2. Build a table to display the sales data for each product category. Label the table elements.
3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.
4. Create a pie chart to represent the distribution of sales across product categories. Include labels.

11. Dataset: Geographic Data

City	Population	Avg. Temperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.
2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.
3. Build a table to display the geographic data for each city. Label the table elements.
4. Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

Set 12: Product Inventory Management

Dataset: Product Inventory

Product ID	Product Name	Quantity Available	Price (in \$)
1	Product A	250	20
2	Product B	175	15
3	Product C	300	18

1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different product categories.
3. Build a table to show the inventory data for each product. Label the table elements.
4. Develop a Tableau dashboard combining the bar chart, stacked bar chart, and the table for interactive exploration of inventory data.

Set 13: Stock Analysis

Dataset: Stock Prices

Date	Stock A	Stock B	Stock C
2023-01-01	100	150	120
2023-01-02	105	152	118
2023-01-03	110	148	122

1. Create a line chart to visualize the stock prices of three companies (Stock A, Stock B, and Stock C) over a specific time period. Label the axes and title the chart.
2. Generate a bar chart showing the daily percentage change in stock prices for Stock A. Label the chart elements.
3. Build a table to display the stock price data for each company over the given period. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of stock prices.

Set 14

Dataset: Energy_Consumption_Data.csv

Sector	Region	Month	Temperature (°C)	Units_Consumed (kWh)	Cost ()	Renewable_Usage (%)	Peak_Hours
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

1. Import the dataset in **R**. Plot a histogram and density plot for **Units_Consumed**. Compare consumption patterns.
2. Using **R programming**, create a scatter plot between **Temperature** and **Units_Consumed**. Represent **Peak_Hours** using bubble size. Apply transparency.
3. Using **R**, calculate average **Renewable_Usage** for each **Sector**. Create a bar chart. Interpret renewable energy adoption.
4. Import **Energy_Consumption_Data.csv** into **Tableau**. Create a line chart for monthly energy usage trends. Create a heatmap (**Sector vs Region vs Units**). Add filters for **Month** and **Sector**. Design an interactive dashboard.

SET 15– Student Mini Data Analysis

Dataset: Student_Mini_Data.csv

Student_ID	Gender	Study_Hours	Attendance	Math_Score	Science_Score	Exam_Date
S01	Male	2.0	78	62	65	2025-01-10
S02	Female	3.5	90	80	85	2025-01-10
S03	Male	1.5	70	55	58	2025-02-12
S04	Female	4.0	95	90	92	2025-02-12
S05	Male	2.8	85	72	74	2025-03-15
S06	Female	3.0	92	82	86	2025-03-15

Questions

1. Import the dataset in **R**. Plot a histogram for **Math_Score**. Draw a boxplot of **Science_Score** by **Gender**. Add proper title and labels. Interpret the results.
2. Create a scatter plot using **R programming** between **Study_Hours** and **Math_Score**. Use different colors for **Gender**. Add a regression line. Explain the relationship between study time and performance.
3. Using **R**, convert **Exam_Date** into Date format. Calculate monthly average Math scores. Plot a line chart for the trend. Apply moving average smoothing. Identify any increasing or decreasing pattern.
4. Import **Student_Mini_Data.csv** into **Tableau**. Create a bar chart showing average Math scores by Gender. Create a line chart for score trends over time. Add a filter for **Exam_Date**. Design a simple interactive dashboard.

Set 16. Patient Health Risk Analysis

Dataset

Patient_ID	Age	BMI	BP	Cholesterol
P1	25	22	120	180
P2	40	28	135	210
P3	55	30	145	240
P4	35	26	130	200
P5	60	32	150	260

Questions

- Using R, create a scatterplot matrix for **Age, BMI, BP, and Cholesterol**. Examine pairwise relationships to identify strong correlations, clusters, or potential risk patterns.
- Construct a Q-Q plot and ECDF for **Cholesterol** levels to assess distribution behaviour. Comment on normality assumption and skewness.
- Using R, create a bar chart showing the average values of each health indicator (**Age, BMI, BP, Cholesterol**). Include axis labels, a descriptive title, and distinct colors for each bar. How do these averages relate to general patient health risk?
- Create an interactive Tableau dashboard with a heatmap highlighting high-risk patients and an Age filter for exploring risk patterns. Include hover tooltips showing all health indicators. Analyze which patients are highest risk, how age affects risk, and how this dashboard can aid healthcare professionals in prioritizing care.

SET 17 – Vehicle Performance Analysis

Dataset

Vehicle_ID	Engine_Size (L)	Horsepower	Fuel_Efficiency (km/l)	Top_Speed (km/h)	Safety_Rating
V1	1.5	110	18	180	4
V2	2.0	150	15	200	5
V3	3.0	250	12	250	5
V4	2.5	200	14	220	4
V5	1.8	130	17	190	3

Questions

- Using R, develop a violin plot to compare the distribution of **Fuel_Efficiency** across different **Safety_Rating** categories. Examine the spread, central tendency, and variability within each group, and interpret what this indicates about efficiency patterns.

- Using R, construct a scatter plot to explore the relationship between **Horsepower** and **Top_Speed**, applying color differentiation based on **Engine_Size**. Analyze the strength and direction of correlation and find observable performance trends.
- Using R, create a correlation heatmap for all numerical variables in the dataset. Identify the variables most strongly associated with **Top_Speed** and their influence on overall vehicle performance.
- In Tableau, design an interactive dashboard incorporating bar charts and scatter plots, along with filters for **Engine_Size** and **Safety_Rating**, to highlight vehicles that balance fuel efficiency and high performance. Provide analytical interpretation of the insights derived from the dashboard.

SET 18 – Student Academic Performance

Dataset

Student_ID	Age	Study_Hours	Attendance (%)	Test_Score	Participation_Score
S1	19	12	90	85	8
S2	21	8	70	70	7
S3	20	15	95	92	9
S4	22	10	85	80	8
S5	23	7	60	65	6

Questions

- Using R, create a stacked area chart showing **Test_Score** and **Participation_Score** across students. Analyze trends.
- Using R, generate a boxplot of **Study_Hours** grouped by **Attendance** quartiles. Interpret patterns. Include axis labels, title, and colors for each quartile. Analyze the boxplot to identify median and interquartile range of study hours per attendance quartile and any outliers in study hours.
- Using R, construct a density plot for **Test_Score** to evaluate score distribution and outliers.
- In Tableau, design an interactive scatterplot with **Study_Hours** on X-axis, **Test_Score** on Y-axis, colored by **Participation_Score**. Include filters for **Age**.

SET 19 – Mobile App Usage Analysis

Dataset: Mobile_App_Usage.csv

User_ID	Gender	Age	Screen_Time (hrs)	App_Usage_Count	Data_Used (GB)	Satisfaction	Usage_Date
U01	Male	20	4.5	18	2.4	3	2025-01-08
U02	Female	22	6.0	25	3.8	5	2025-01-08
U03	Male	19	3.2	12	1.6	3	2025-02-11
U04	Female	21	7.1	30	4.5	5	2025-02-11
U05	Male	23	2.8	10	1.2	2	2025-03-14
U06	Female	20	5.4	22	3.1	4	2025-03-14

Questions

1. Import the dataset in R. Plot a histogram and density plot for **Screen_Time**. Add suitable colors and labels. Interpret the screen-time distribution.
2. Create a scatter plot using R programming between **Data_Used** and **Screen_Time**. Calculate the correlation between them. Add a trend line. Explain the relationship.
3. Using R, find the average **Satisfaction** score for each **Gender**. Create a bar chart for the averages. Add value labels. Interpret user satisfaction patterns.
4. Import **Mobile_App_Usage.csv** into Tableau. Create a bubble chart (**Screen_Time vs Data_Used**). Include filters for **Gender** and **Usage_Date**. Design an interactive dashboard and summarize the key insights.

SET 20. Hospital Appointment Analysis

Dataset: Hospital Appointments

Patient ID	Age	Waiting Time (Days)	Appointment Status
1	25	2	Attended
2	40	5	Missed
3	35	1	Attended
4	50	7	Missed
5	29	3	Attended

Questions

1. Create a histogram showing patient ages.
2. Generate a pie chart for appointment status distribution.
3. Build a bar chart of waiting times by patient.
4. Create a scatter plot of age vs waiting time.

SET 21. Movie Ratings Analysis

Dataset: Movie Ratings

Movie ID	Genre	Rating	Duration (Min)
1	Action	4.5	120
2	Comedy	3.8	90
3	Drama	4.2	140
4	Action	4.7	130
5	Comedy	3.5	95

Questions

- 5. Create a histogram of movie ratings.
- 6. Generate a pie chart showing genre distribution.
- 7. Build a bar chart of average ratings by genre.
- 8. Create a scatter plot of duration vs rating.

SET 22. Library Borrowing Records

Dataset: Library Usage

User ID	Books Borrowed	Days Kept	Fine Amount
1	2	10	0
2	5	25	15
3	3	14	0
4	6	30	20
5	1	7	0

Questions

- 9. Create a histogram of books borrowed.
- 10. Generate a pie chart for users with and without fines.
- 11. Build a bar chart of fine amounts by user.
- 12. Create a scatter plot of days kept vs fine amount.

SET 23. Airline Passenger Analysis

Dataset: Airline Passengers

Passenger ID	Age	Flight Hours	Satisfaction
1	28	2	High
2	45	8	Medium
3	33	5	High
4	52	10	Low
5	39	6	Medium

Questions

- 13. Create a histogram of passenger ages.
- 14. Generate a pie chart of satisfaction levels.
- 15. Build a bar chart of flight hours by passenger.
- 16. Create a scatter plot of age vs flight hours.

SET 24. Hotel Booking Analysis

Dataset: Hotel Bookings

Booking ID	Stay Nights	Guests	Room Type
1	2	2	Standard
2	5	4	Deluxe
3	3	1	Standard
4	7	3	Suite
5	4	2	Deluxe

Questions

- 17. Create a histogram of stay nights.
- 18. Generate a pie chart of room types.
- 19. Build a bar chart of guests per booking.
- 20. Create a scatter plot of guests vs stay nights.

SET 25. Fitness Tracker Analysis

Dataset: Daily Fitness Activity

User ID	Steps	Calories Burned	Active Minutes
1	7000	250	40
2	10000	400	60
3	8500	320	50
4	12000	500	75
5	6500	220	35

Questions

- 21. Create a histogram of daily steps.
- 22. Generate a pie chart of activity level categories.
- 23. Build a bar chart of calories burned by user.
- 24. Create a scatter plot of steps vs calories burned.

SET 26. Restaurant Orders Analysis

Dataset: Restaurant Orders

Order ID	Items Ordered	Bill Amount	Dining Type
1	2	25	Dine-In
2	5	60	Takeaway
3	3	35	Dine-In
4	4	50	Delivery
5	2	20	Takeaway

Questions

- 25. Create a histogram of bill amounts.
- 26. Generate a pie chart of dining types.
- 27. Build a bar chart of items ordered by order.
- 28. Create a scatter plot of items ordered vs bill amount.

SET 27. Energy Plant Monitoring

Dataset: Power Plant Operations

Plant ID	Output (MW)	Temperature	Status
1	120	65	Active
2	150	70	Active
3	100	60	Maintenance
4	170	75	Active
5	110	62	Maintenance

Questions

- 29. Create a histogram of power output.
- 30. Generate a pie chart of plant status.
- 31. Build a bar chart of temperatures by plant.
- 32. Create a scatter plot of output vs temperature.

SET 28. Social Media Engagement Analysis

Dataset: Social Media Posts

Post ID	Likes	Comments	Shares
1	120	15	10
2	200	30	20
3	150	20	12
4	300	40	35
5	180	25	18

Questions

- 33. Create a histogram of likes.
- 34. Generate a pie chart of total engagement components.
- 35. Build a bar chart of comments by post.
- 36. Create a scatter plot of likes vs shares.

SET 29. Water Quality Monitoring

Dataset: Water Samples

Sample ID	pH Level	Turbidity	Quality
1	7.2	3	Good
2	6.8	5	Fair
3	7.5	2	Good
4	6.5	7	Poor
5	7.1	4	Fair

Questions

- 37. Create a histogram of pH levels.
- 38. Generate a pie chart of water quality categories.
- 39. Build a bar chart of turbidity by sample.
- 40. Create a scatter plot of pH level vs turbidity.

SET 30. Music Streaming Analysis

Dataset: Music Streaming

Song ID	Duration (Min)	Streams (000s)	Genre
1	3.5	150	Pop
2	4.2	200	Rock
3	3.8	180	Pop
4	5.0	250	Hip-Hop
5	4.1	170	Rock

Questions

1. Create a histogram of song durations.
2. Generate a pie chart of genre distribution.
3. Build a bar chart of average streams by genre.
4. Create a scatter plot of duration vs streams.

